

R464: Inner Bridges

TFPT RH programme

30 August 2026

Claim boundary. This note proves one finite matrix implication and transcribes one production coefficient family. It does not prove the remaining channel representation, the Gauss–Mellin–digamma identity, or the Riemann hypothesis.

1 Finite PSD bridge

The exact proved Lean theorem is:

```
selectedACapPsdImpliesPlainReads_of_representation  
(hrep : SelectedReadQuadraticRepresentation) :  
SelectedACapPsdImpliesPlainReads.
```

Its proof is the finite implication

$$A_{\text{cap}} \succeq 0, \quad \text{fullRead}(f) = z^{\mathsf{T}} A_{\text{cap}} z \implies \text{fullRead}(f) \geq 0.$$

The sole remaining channel proposition is `SelectedReadQuadraticRepresentation`. It asks for the actual augmented direction z , including Fourier/Chebyshev coordinates, whitening, border coordinate, and pole correction.

The old target has not been weakened. In particular, it quantifies over all `GridElements` satisfying only `f.meshExp <= selectedMesh k`. Their `steps` field is unbounded, whereas A_{cap} is finite-dimensional. At $k = 5$, the probe records the admissible arithmetic witness

$$\text{meshExp} = 19, \quad \text{steps} = 11, \quad \text{cap} = 10.$$

This is a quantifier obstruction to a degree- $<\text{cap}$ argument, not a numerical counterexample to positivity.

2 Archimedean bridge

`productionArchLag` is now a concrete Lean definition. For $i = 0$ it is the Euler- $\log \pi$ regularized near-cell integral `productionArchLagNear`; for $i \geq 1$ it is the ordinary tent integral `productionArchLagFar`. The definitions transcribe Python `arch_A_near`, `arch_A_far`, and `arch_lags`.

The former generic stabilization hole is replaced one-for-one by the single named theorem

```
arch_gauss_mellin_digamma_identity :  
ArchGaussMellinDigammaIdentity.
```

The existing theorem `arch_elementwise_stabilization` is now a direct application of this identity. Mathlib 4.29.1 contains the digamma functional equation and special values, but its source explicitly lists Gauss’s integral representation as a TODO; the needed Mellin identification is therefore not available from the imported API.

3 Numerical seal and census

The smoke probe pins $(c_0, c_1, c_{\text{last}}, \sum_i c_i)$:

k	c_0	c_1	c_{last}	$\sum_i c_i$
5	0.294628394551	-0.735655442926	-0.033473428435	-2.169368995630
9	1.009256397265	-0.714628002714	-0.003999307222	-2.091132571876
10	1.009256397265	-0.714628002714	-0.002827927599	-2.117589931320

The Lean sorry census remains 8: three in `Canonical.lean`, one named Gauss–Mellin–digamma identity in `Elementwise.lean`, three external arrows in `ExternalBridges.lean`, and one source theorem in `Source.lean`.

4 Kill table

Candidate	Status	Reason
PSD alone closes all reads	killed	Unbounded steps ; no cap-space direction has been constructed.
Opaque production arch lags	killed	Near and far coefficient integrals are concrete Lean definitions.
Current Mathlib closes the arch seam	killed	Gauss’s digamma integral is an explicit Mathlib TODO; Mellin identification is absent.
R464 proves RH	killed	Two internal named remainders and three external bridges remain.